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REVIEW PAPER

Effects of environmental variables on fish feeding ecology: implications for the performance of baited fishing gear and stock assessment

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The effectiveness of baited fishing gear ultimately depends upon behaviour of the target species – activity rhythms, feeding motivation, and sensory and locomotory abilities. While any environmental parameter that mediates feeding or locomotion can have an important influence on the active space presented by the bait and fish catchability, few biologists have considered how such variation in behaviour might affect catch per unit effort (CPUE) and the resultant stock abundance estimates or population parameters. This review reveals that environment-related variation in feeding behaviour can act through four different mechanisms: metabolic processes, sensory limitations, social interactions and direct impacts. Water temperature, light level, current velocity and ambient prey density are likely to have largest effects on fish catchability, potentially affecting variation in CPUE by a factor of ten. Feeding behaviour is also density-dependent, with both positive and negative effects. Over time and geographic space a target species can occupy wide ranges of environmental conditions, and in certain cases, spatial and temporal variation in feeding biology could have a larger impact on CPUE than patterns of abundance. Temperature, light and current can be measured with relative facility and corrections to stock assessment models are feasible. Making corrections for biological variables such as prey density and bait competitors will be more difficult because the measurements are often not practical and relationships to feeding catchability are more complex and poorly understood. There is a critical need for greater understanding of how environmental variables affect feeding-related performance of baited fishing gear. A combination of field observations and laboratory experiments will be necessary to parameterize stock assessment models that are improved to accommodate variation in fish behaviour. Otherwise, survey data could reveal more about variation in behaviour than abundance trends. © 2004 The Fisheries Society of the British Isles

Key words: active space; bait; CPUE; feeding biology; feeding motivation; stock assessment.

'As in all other forms of fishing, using a line and baited hook involves a degree of luck, but the odds are shortened by careful study of the behaviour and likely distribution of the species to be caught.' C.S. Wardle, 1993

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INTRODUCTION

Stock assessment surveys using baited fishing gear assume that catch per unit effort (CPUE) is proportional to overall population abundance. Punsly & Nakano (1992), however, pointed out that behaviour of the target species in response to environmental variables such as thermal structure, dissolved oxygen and prey availability, could have greater effects on CPUE than fish abundance. Variation in CPUE is affected by two factors – fish availability to the gear (*i.e.* distribution over time and space) and vulnerability to the gear (*i.e.* catchability) (Engås & Løkkeborg, 1994; Arreguín-Sánchez, 1996). Considerable progress has been made with respect to fish availability in stock assessment models for pelagic fishes based upon longline catches. For example, by correcting survey data for the number of hooks that are within the preferred depth range for a target species, the issue of effective *v.* nominal effort can be addressed, at least in part (Hinton & Nakano, 1996; Bigelow *et al.*, 2002).

Differential vulnerability to baited gear results from the effects of environmental variables on fish activity, feeding motivation and the ability to detect, locate and consume baits. First, the area sampled by a longline or fish pot is dependent upon the 'active space' over which feeding attractants from the baits are present in concentrations above the response threshold for the fish. This active space is determined by the release rate and transport of chemical cues through the water as well as the chemosensory threshold for an individual at a specific time (Løkkeborg *et al.*, 1995). Each of these components is highly variable. Second, only a certain proportion of fish within an active space respond to baited gear, and only some of those are motivated to strike baits or enter fish pots and be caught (Furevik, 1994; Løkkeborg, 1994; Kaimmer, 1999; Stoner, 2003). Hence, fish catchability can vary with time, location, and any environmental variables, including the presence of conspecifics and other competitors that influence feeding behaviour during each gear set.

Successful fishing operations and stock assessments with baited gear depend upon fundamental knowledge of fish behaviour, including changes in distribution, patterns of activity and feeding motivation. Great emphasis has been placed upon designing fishing methods and equipment to exploit known behaviour patterns, with goals of optimizing fishing locations, attractiveness of baits and selection for particular target species and fish sizes. For example, knowledge of diurnal, tidal and seasonal variation in activity and natural feeding patterns is used to optimize the timing of fishing activities (Johannes, 1981; von Brandt, 1984; Engås & Løkkeborg, 1994). Numerous field trials have been conducted to examine the role of hook and trap types, different baits, and spacing of hooks and fish pots on longlines (Myhre, 1969; Johannessen *et al.*, 1993; Furevik & Løkkeborg, 1994; Woll *et al.*, 2001; and reviews by Løkkeborg & Bjordal, 1992; Engås & Løkkeborg, 1994; Furevik, 1994). Also, models have been developed to accommodate variations in the transport and decomposition of olfactory cues and resultant changes in active space (Murphy, 1960; Eggers *et al.*, 1982; McQuinn *et al.*, 1988; Løkkeborg *et al.*, 1995; Sigler, 2000). The potential importance of environmental variables such as light level, temperature, density and size distribution of competitors, and abundance of natural prey have also been mentioned for potential impact on performance of baited fishing

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gear (Quinn *et al.*, 1985; Engås & Løkkeborg, 1994; Sigler, 2000). In fact, any environmental variable that affects activity, feeding motivation, or ability to detect, locate and attack a bait will have an impact on active space and fish catchability, but few experiments on these subjects have been carried out, and a ranking of importance for these factors is not available.

Given that activity and feeding are likely to be influenced by a large number of environmental variables, the question needs to be asked to what degree do bait-dependent surveys and stock assessments reveal spatial and temporal patterns in the abundance of fish or simply the effects of environmental variables on fish catchability? This question is particularly crucial with growing evidence that populations of large predatory fishes, often exploited with baited fishing gear, are rapidly declining (Myers & Worm, 2003). The purpose of this review is to identify, evaluate and rank environmental variables with respect to their likely influence on vulnerability of fishes to baited fishing gear and to determine whether or not improvements to stock assessments can be made by correcting for environmental variation. How some of these variables affect availability of fish to fishing gear has been discussed by others (Gunderson, 1993; Engås & Løkkeborg, 1994; Hinton & Nakano, 1996; Michalsen *et al.*, 1996) and is beyond the scope of this paper. Also, transport of bait odour and natural rhythmic patterns of activity and feeding are not discussed here. Rather, emphasis is placed on physical and biological conditions in the environment that, independent from natural temporal rhythms, affect changes in activity and locomotion, feeding motivation, bait scent detection, searching behaviour, bait location and probabilities of bait attack by fishes. In many cases reference is made to the basic literature on fish feeding because studies specific to bait performance have not been conducted. Also, emphasis is placed on experimental work conducted with baits and baited hooks, and less with fish traps which involve more complex behavioural mechanisms than hook gear. This analysis, however, will have direct relevance to fishing with all forms of baited gear including traditional angling and trolling gear, demersal and pelagic longlines, and fish pots. This review is intended to point out the most serious potential limitations in standard bait-dependent stock assessments so that the basic research needed to improve stock assessment models can be identified and pursued.

ENVIRONMENTAL CONTROL OF FEEDING BEHAVIOUR

TEMPERATURE

Temporal and spatial variation in water temperature is one of the most pervasive and ecologically relevant parameters in marine systems. Temperature is the most important factor governing metabolism in ectothermic fishes (Fry, 1971), and even within a normal range of physiological tolerance, temperature can have significant effects on activity, swim speed, feeding and other fundamental behaviours relevant to bait-dependent fishing. [In this review the influence of temperature is discussed with respect to ectothermic fishes and not temperature-regulating pelagic species such as tunas and billfishes.]

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In most fishes, swim speed will increase with temperature up to some maximum, then decrease precipitously when thermal stress begins, as shown in the seminal research on temperature and metabolism in sockeye salmon (*Oncorhynchus nerka* Walbaum) (Brett, 1971). Since that early investigation, numerous others have reported temperature-related variation in spontaneous locomotor activity and swim speeds for economically significant species including lake whitefish (*Coregonus clupeaformis* Mitchill) (Scherer & Harrison, 1988), Atlantic cod (*Gadus morhua* L.) (Castonguay & Cyr, 1998), sea bass (*Dicentrarchus labrax* L.) (Koumoundouros *et al.*, 2002), flatfishes (He, 2003) and sablefish (*Anoplopoma fimbria* Pallas) (Stoner & Sturm, 2004). Decreased activity and swim speed result in reductions in the probability of encountering the odour plume from a bait, the probability of locating a bait, and hence the vulnerability to baited fishing gear.

Temperature can also have subtle effects on taste preferences in fishes. For example, stimulatory efficiencies of taste receptors for different amino acids in grass carp (*Ctenopharyngoden idella* Cuvier & Valenciennes) and stellate sturgeon (*Acipenser stellatus* Pallas) are known to vary with temperature, and probably affect observed seasonal shifts in diets (Kasumyan & Døving, 2003). Changes in receptiveness such as these are likely to affect responses to baits.

Feeding motivation is determined jointly by the metabolic debt (systemic need) of the fish and the amount of food in the stomach (gastric volume), and these two factors are affected by many physiological and ecological factors (Colgan, 1973, 1986). Because temperature affects metabolic rates and gastric evacuation in ectothermic fishes (He & Wurtsbaugh, 1993; Andersen, 2001; Angilletta *et al.*, 2002), hunger, search for food and feeding intensity can all be closely related to temperature. [In this review 'hunger' refers specifically to the physiologically-induced need for food caused by food deprivation or high metabolism. The term 'feeding motivation' is used more generally to describe the desire to feed regardless of mechanism. This would include propensity to feed even when satiated, for example in the context of social facilitation. Conversely, feeding motivation could be inhibited in a hungry individual by the presence of a threat to safety.] Food intake normally increases steadily with temperature up to some maximum, then declines rapidly (Brett, 1979). Kestemont & Baras (2001) list temperatures at which maximum food consumptions occur for a variety of fish species in culture. There are now many examples of temperature effects on feeding behaviour and metabolism in species significant to commercial fisheries, particularly flatfishes (Peters & Boyd, 1972; Malloy & Targett, 1991, 1994; Fonds *et al.*, 1992; Madon, 2002), Atlantic cod (Jobling, 1988) and salmonids (Nicieza & Metcalfe, 1997; Forseth *et al.*, 2001).

In most cases, food consumption will increase with temperature, even past levels that support maximum growth (Jobling, 1997). In turbot (*Scopthalmus maximus* L.) feeding demand is directly proportional to metabolic rate, which increased exponentially between 6 and 22° C, well above optimum temperature for the species (Mallekh & Lagardère, 2002). Rates of change in temperature may also affect feeding intensity (Kestemont & Baras, 2001), with rapid changes often inhibiting feeding. Acclimatization to declining temperature in salmon and trout can take several weeks (Koskela *et al.*, 1997).

Temperature can also affect feeding periodicity in fishes. For example, Atlantic salmon (*Salmo salar* L.) in water temperatures >6° C feed on a diurnal

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schedule, but when temperatures are <6° C, 90% of feeding is nocturnal (Fraser *et al.*, 1993, 1995). A temperature-compensated circadian clock affecting locomotion also has been observed in zebrafish (*Danio rerio* Hamilton-Buchanan) (Hurd *et al.*, 1998). Consequently, fish catchability and the effectiveness of baited fishing gear can be strongly affected by temperature through a variety of mechanisms.

In his analysis of abundance estimation for sablefish using longline gear Sigler (2000) suggested that 'the weak link between catch rate and abundance is the unknown extent that factors such as temperature and food availability affect the proportion of fish caught.'. Addressing this concern, Stoner & Sturm (2004) conducted controlled experiments to examine responsiveness to baits in large sablefish acclimated to a range of temperatures (2–8° C) spanning the normal range for the species. Sablefish held in low temperature had low swim speeds, reduced responses to olfactory cues, they attacked and consumed fewer baits, the time taken to attack and consume baits was greatly increased, and handling times increased. Baits were frequently passed over in low temperatures and attacks were often tentative compared with aggressive attacks observed in high temperature. These results are all relevant to sablefish catch in the field. When baited traps were set for sablefish on the continental slope off Oregon in 2001 there was an inverse relationship between sablefish catch rate and temperature (1·6 to 6° C) (M. Schirripa, pers. comm.). It is possible that distribution and abundance was directly associated with spatial patterns in temperature. Given the laboratory findings, however, it is equally plausible that catch rates in the traps were related to temperature-mediated variation in sablefish locomotion, bait detection and feeding motivation. While care is needed in extrapolating the results of laboratory experiments directly to the field (Løkkeborg *et al.*, 1993), Stoner & Sturm (2004) concluded that catchability coefficients need to span at least an order of magnitude within the natural temperature range of sablefish.

LIGHT LEVEL

Intensity of light at the water surface varies with time of day and season, depending upon atmospheric conditions, while light level below the surface is a function of surface illumination, wave action, water clarity and depth. The result is that fishes experience large variation in light environment on both temporal and spatial scales. Diel periodicities in activity and feeding are generally well-known for fishes with high economic value, and are not the primary subject of this discussion. Light level can also have direct effects on non-rhythmic behaviour patterns that cause variation in catch rates with baited fishing gear. For example, activity, swim speeds and feeding propensity can vary with light level because light affects vision and contrast.

There is little doubt that light level has a direct effect on locomotion in fishes, independent of diel rhythms. General activity in lake whitefish is strongly affected by light level as well as water temperature (Scherer & Harrison, 1988), and similar relationships have been observed for brook trout (*Salvelinus fontinalis* Mitchill) (Marchand *et al.*, 2003). In fact, light intensity is used routinely to influence activity and feeding of fishes held for aquaculture (Trippel

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& Neil, 2003). More directly relevant to bait function, Stoner (2003) conducted experiments with Pacific halibut (*Hippoglossus stenolepis* Schmidt) using infrared lighting in large laboratory pools to make video recordings at four different light levels including total darkness. General activity was dependent upon light level, with highest activity occurring in full sunlight conditions. Light level had no effect on the time to olfactory detection of baits and first upstream movement, but times to locate and attack squid baits increased significantly in darkness. Only 17% of the baits presented in darkness were consumed while 70–100% of the baits were consumed at higher light levels. Similarly, Atlantic cod (*Gadus morhua*) respond to bait odour both day and night and are able to locate bait at night, but more slowly (Løkkeborg & Fernö, 1999).

In most cases, fish are attracted to baits by chemical cues, but vision is important in near-field location and final attack on a food item. The light threshold for feeding was high (7×10^{-3} μmoles photons $m^{-2} s^{-1}$) in the dwarf scorpion fish (*Scorpaena papillosus* Schneider & Forster), which lives in shallow reef habitats (Montgomery & Hamilton, 1997). In contrast, many economically important fishes, especially those living in turbid water or in deep shelf environment, have low light thresholds for visual feeding. For example, McMahon & Holanov (1995) found thresholds near 2×10^{-4} lx ($3{\cdot}9 \times 10^{-6}$ μmoles photons $m^{-2} s^{-1}$) for piscivory in largemouth bass (*Micropterus salmoides* Lacépède). Thresholds for planktivorous juvenile walleye pollock (*Theragra chalcogramma* Pallas) and sablefish were 5×10^{-7} and 5×10^{-5} μmoles photons $m^{-2} s^{-1}$, respectively (Ryer & Olla, 1999). The light threshold for juvenile Pacific halibut to locate and attack baits was *c.* 10^{-6} μmoles photons $m^{-2} s^{-1}$ (Stoner, 2003). This corresponds with a depth of 200–250 m in clear-water conditions on a clear sunlit day, but only 60 m in turbid coastal waters. Because of large variation in cloud cover, water transparency and ocean surface conditions, light levels at depth will be highly variable over a wide range of temporal and spatial scales. Laboratory results show that Pacific halibut can locate and consume baits in darkness without visual cues, but the attacks are fewer, less efficient and probably depend upon olfaction. Bjordal & Løkkeborg (1996) drew similar conclusions for Atlantic cod, which are captured on longlines in the Barents Sea in winter. The result, however, is that fish catchability will be significantly lower when and where light level is below the visual threshold for efficient bait location and attacks.

TURBIDITY

The effects of turbidity created by suspended sediment or other particulate materials on activity and feeding by fishes may be analogous to the effects of ambient light. There is abundant literature on the role of light and turbidity on prey capture by planktivorous fishes (Vinyard & O'Brien, 1976; Gardner, 1981; Macy *et al.*, 1998), and turbidity is known to reduce capture efficiency in piscivorous fishes (Gregory & Levings, 1998; Beauchamp *et al.*, 1999; Reid *et al.*, 1999). Turbidity reduces and scatters light and can reduce apparent contrast, and feeding impairment appears to act primarily through reduced reactive distances (Vogel & Beauchamp, 1999; Sweka & Hartman, 2001). The effects of turbidity, however, can be different in different fish species (Benfield

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& Minello, 1996; Rowe & Dean, 1998; DeRobertis *et al.*, 2003) depending upon their sensory capabilities and to what extent the fish depends upon visual cues for feeding, similar to the effects of reduced light. For example, Rowe *et al.* (2003) found that turbidity produced little impairment in feeding by rainbow trout (*Oncorhynchus mykiss* Walbaum) because the species can feed effectively with non-visual cues. Similarly, for the marine species such as sablefish that normally function in low light conditions and locate baits through olfaction (Løkkeborg *et al.*, 1995), turbidity may have relatively little impact on feeding. In extreme conditions of turbidity, particularly where organic-rich sediments are suspended, however, chemical cues associated with baits might be masked.

It is possible that turbidity could influence the general activity and feeding motivation of fishes living in lighted environments through either direct effects or indirectly through reduction of light level. These effects are little investigated in marine fishes, but experiments with six diadromous fishes show that reductions in feeding associated with turbidity resulted from reduced feeding abilities and not reduced feeding motivation or willingness to search (Rowe & Dean, 1998).

Localized sediment clouds often reveal prey or biological activities that expose prey to fish predators, and many species respond to this kind of cue. Consequently, turbidity created by fixed gear striking or moving on the bottom might attract fishes to baited gear when the fish are within visual range of the effects.

CURRENTS AND TIDE

Current velocity is relevant to fishing with baited gear in two different ways. First, olfactory cues that attract fish to baits are transported by water movement. Much attention has been given to this variable because of its central importance in estimating active space for the bait (Eggers *et al.*, 1982; Engås & Løkkeborg, 1994). Second, currents can have direct impacts on the activity and feeding behaviour of fishes, independent of food cues. Anglers, in particular, are aware of variation in catch rates associated with current and tidal phase in a variety of species, but empirical studies are relatively few. Fernö *et al.* (1986) reported that higher activity toward bait is observed in whiting (*Merlangius merlangus* L.) during periods of current than non-current. It is often impossible, however, to separate the positive effect of olfactory cue transport from direct responses to current velocity. Activity in Atlantic cod and haddock (*Melanogrammus aeglefinus* L.) is modified by tidal current (Løkkeborg *et al.*, 1989; Arnold *et al.*, 1994), and both cod and plaice (*Pleuronectes platessa* L.) take up sheltering behaviour under high current conditions (Gerstner, 1998; Gerstner & Webb, 1998). Recent observations of juvenile Chinook salmon (*Oncorhynchus tshawytscha* Walbaum) and steelhead trout (*O. mykiss* Walbaum) show that both species are active day and night under conditions of low river flow, but when river flow is high the fishes are nocturnal and feed only at dusk (Bradford & Higgins, 2001). Tidal currents also affect the vertical distribution of cod and haddock, and their susceptibility to trawl gear (Michalsen *et al.*, 1996). Current-associated variation in fish distribution, general activity and feeding aggressiveness will result in changes in bait location, bait attacks and hooking frequency.

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Tides and tidal currents can be a significant influence even in deep water. Large variation in current strength and direction was observed over tidal periods when current meters were deployed with baited traps for sablefish at 400 m depth on the continental slope off Oregon (A.W. Stoner, unpubl. data). Consequently, tidal and other forms of current may need to be considered for their impact on the performance of baited fishing gear, both in terms of chemical cue transport and direct effects on feeding behaviour.

BAROMETRIC PRESSURE, WIND AND TURBULENCE

The fishing community sometimes attributes variation in catch rates to changes in barometric pressure. No controlled experiments have been conducted and it is difficult to distinguish the effects of pressure from other atmospheric conditions such as cloud cover (*i.e.* light level), wind, turbulence and other variables associated with barometric changes. Several correlative studies in freshwater environment, however, suggest that fishes can respond to barometric pressure. In a study of spawning behaviour in rainbow trout, Peterson (1972) found that barometric pressure was a better predictor of migration than water temperature or light level. Similarly, activity level in black crappie (*Pomoxis nigromaculatus* Lesueur) fitted with ultrasonic transmitters was positively correlated with barometric pressure and not water temperature, water transparency, cloud cover, wind direction, wind speed or precipitation (Guy *et al.*, 1992). Barometric pressure also appears to control depth distribution in sauger (*Stizostedion canadense* Smith) (Jeffrey & Edds, 1999). Because barometric pressure can influence activity and distribution patterns in fishes, impacts on feeding behaviour are also likely. Relevance to marine fishes without swim bladders such as halibut and sablefish awaits more information but relationships have been postulated by fishers.

Turbulence generated by wind sheer and ocean swell can also generate variation in fish feeding. There is extensive literature on the ability of larval fishes to feed in turbulent flow (Lough & Mountain, 1996; Dower *et al.*, 1997; MacKenzie & Kiørboe, 2000). Effects of wind- or swell-induced turbulence on feeding can also be expected for older fishes in shallow water. For example, video observations of *Notothenia coriiceps* Richardson revealed that activity and feeding in the demersal ambush predator were unaffected by tide but greatly reduced when wind speed exceeded *c.* $8\,\mathrm{m\,s^{-1}}$ (North, 1996). Also, large reef-dwelling fishes often suspend feeding during periods of heavy swell (A.W. Stoner, pers. obs.). Turbulence frequently reduces water clarity, further affecting feeding behaviour and bait location.

BOTTOM TYPE

Characteristics of the benthic environment have broad, widely recognized effects on the distribution and abundance of demersal fishes and their prey. Bottom type, particularly vertical structure or rugosity, can also have a direct effect on the ability of fishes to locate prey and feed (Stoner, 1982; Minello & Zimmerman, 1983; Eklöv, 1997; Buckel & Stoner, 2000). Most of the experimental studies of feeding in structured habitats have been conducted with small fishes and live prey. Recent experiments with relatively large summer flounder

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(*Paralichthys dentatus* L.) (Manderson *et al.*, 2000) and Pacific halibut (Ryer *et al.*, 2004), however, show that prey consumption is adversely affected by the reduced visual field imposed by complex habitats. The effect on bait location may be less than that with live prey; however, Ryer *et al.* (2004) observed that general activity in Pacific halibut is significantly less in a complex habitat compared with open sand. Habitat rugosity may also inhibit transport of olfactory cues over the bottom and make flows more complex.

AVAILABLE PREY

The density and composition of prey is highly variable over time and space because of natural reproductive cycles, variable recruitment, migration patterns and changing predation pressures on the prey populations. The prey field can have several different effects on fishes captured with baited gear: 1) Abundance of suitable prey can affect hunger level and feeding motivation and 2) abundance of chemical cues from prey (or other organic compounds) can mask or compete with bait cues.

The effect of prey abundance on predator hunger level is probably the most important mechanism relevant to baited fishing gear because hunger has a large impact on responsiveness to bait odours and feeding motivation. Laboratory experiments conducted with sablefish show that thresholds for responses to chemical cues from bait vary by several orders of magnitude in proportion to hunger level (Løkkeborg *et al.*, 1995). Calculations for the maximum distances over which sablefish would initiate food search behaviour ranged from just 10 m in satiated fish to several kilometers in hungry fish. More recent laboratory experiments show that hunger level had little effect on spontaneous activity of sablefish before bait odour was introduced (Stoner & Sturm, 2004). During the period immediately after introduction of a bait odour, however, general activity and swim speeds increased in proportion to the food deprivation time (1–6 days), while times to locate and attack baits decreased with increased deprivation. Similarly, times to detect, locate, attack and handle baits all decreased with increasing food deprivation (1–4 days) in Pacific halibut (Stoner, 2003). Attack times decreased from hours to minutes, while the percentage of baits located and consumed increased. These recent studies confirm the conclusion that active space will increase as hunger level increases and as chemosensory threshold decreases. Because bait detection abilities, swim speeds and general feeding motivation all decrease with satiation, high prey density can result in both smaller active space for baited gear and reduced fish catchability.

Evidence also exists for the effects of prey density on responsiveness to baited gear in the field. Engås & Løkkeborg (1994) reported low catch rates for cod on longlines during periods when capelin (*Mallotus villosus* Müller) were abundant. The assumption is that the cod were satiated and not attracted to baits. More recently, Bertrand *et al.* (2002) used extensive acoustic survey data to evaluate relationships between hydrographic conditions, prey density, and the distribution and catchability of tunas (*Thunnus* spp.) in the tropical Pacific. Catch rates were particularly high where prey patches were small and had low density. Conversely, catch rates were low where prey patches were large and dense. Various interpretations are possible for low catch rates – either abundant

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natural prey resulted in the tuna being constantly satiated or high-density presence of prey results in a low probability of bait attack. It is also plausible that high ambient concentrations of prey reduce the effectiveness of chemical cues emitted from the baits (see below). The authors concluded that large-scale variation in prey density affects tuna distribution, while small-scale variation affects feeding motivation and attack rates on baits. Because catch rates are a function of both distribution and feeding motivation, prey catch characteristics are important in interpreting pelagic longline catches. Similar mechanisms undoubtedly play an important role in fishing with demersal longlines and fish pots.

Dissolved free amino acids (DFAA) are probably the primary active compounds affecting olfactory responses in fishes (Bardach, 1975; Hara, 1975), and the thresholds for detection and behavioural response to DFAAs have been reported for various marine and freshwater fishes (Johnstone, 1980). As discussed above, active space for baited fishing gear will often increase with fish hunger or with decreasing chemosensory threshold. In hake (*Urophycis chuss* Walbaum) the amount of time spent in search-related swimming increased in direct proportion to the concentration of chemical food cues (Pearson *et al.*, 1980). Johnstone (1980), however, showed that small increases in the background concentration of a DFAA (*i.e.* glycine) resulted in large decreases in sensitivity of cod to amino acid cues. Consequently, ambient concentrations of amino acids and other organic compounds affecting olfactory sensitivity, including those produced by the prey field, could have a significant effect on the active space affected by baits. While attempts have been made to model transport of bait plumes (Eggers *et al.*, 1982; McQuinn *et al.*, 1988; Sigler, 2000), including an evaluation of variable thresholds for bait response in one case (Løkkeborg *et al.*, 1995), the effects of competing ambient compounds have rarely been considered.

COMPETITION AND SOCIAL FACILITATION

Competition for baits on a longline, as with natural prey items, can take either of two basic forms: 1) exploitative competition – when baits are consumed more rapidly or efficiently by certain species, size classes or groups, thus making the baits unavailable to the less efficient feeders (*i.e.* gear saturation) and 2) interference competition – when individuals actively prevent access to the baits by others. Both forms of competition can occur intraspecifically and interspecifically. Attack rate on baits can also be enhanced by the presence of perceived competitors (social facilitation) and, therefore, competition is not always negative with respect to catch rates.

Size structure of the target species in a particular location or time is part of the biological environment, which affects catches on baited gear. Selective capture of large over small individuals on longlines is recognized (Hamley & Skud, 1978; Bertrand, 1988; Engås *et al.*, 1996). This phenomenon has been interpreted in terms of exploitative competition because large fish have higher swimming speeds and can exploit baits more quickly than small fish (Hart, 1993). Recent experiments with Pacific halibut in large laboratory pools show that large individuals are, in fact, dominant over small individuals (50% mass

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differential) in competition for limited numbers of baits, although small fish were more responsive initially (Stoner & Ottmar, 2004). When large and small halibut converged on a single bait, the smaller individual often hesitated to attack the bait, allowing the large fish to win in nearly all encounters, but overt aggression was not observed.

Use of catch per unit effort (CPUE) as an index of stock size depends upon the assumption that catch of non-target species is constant. Both Skud (1978) and Quinn *et al.* (1985), however, speculated that competition for baits with other species probably causes spatial variation in CPUE for Pacific halibut, violating the stock assessment assumption. Capture of unwanted species on longlines is common and bait theft by small fishes and invertebrates is well known. Also, species differ in their abilities to locate baits and willingness to attack. For example, direct observations show that cod have a more intense response to baits than haddock (Løkkeborg *et al.*, 1989). Rothschild (1967) pointed out the problem of bait competition long ago, and noted that spatial variation in abundance of the target species, size structure and competitors for bait could affect CPUE on longlines. He developed a stochastic model to compute CPUE for the target species, which is independent of competition for space on the fishing gear.

Interference competition on longlines is less well documented than exploitative competition. Large Atlantic cod and torsk (*Brosme brosme* L.) have been observed defending baits from smaller individuals (Løkkeborg & Bjordal, 1992), and cod are known to defend baits from whiting (Ferno *et al.*, 1986). Also, Godø *et al.* (1997) observed aggressive bait-defense behaviour by wolffish (*Anarhichas* sp.) over cod and haddock, and speculated that such dominance could affect longline catches and stock assessments for the subordinate species. Recent studies on catchability of American lobsters (*Homarus americanus* H. Milne-Edwards) using underwater cameras led Jury *et al.* (2001) to conclude that agonistic interactions among individuals were the single most important factor influencing their rate of entry into baited traps and that CPUE is not always a good indicator of lobster density.

Presence of bait competitors is not always negative. Several experimental studies show that activity, feeding and growth is socially facilitated in economically significant species including striped mullet (*Mugil cephalus* L.) (Olla & Samet, 1974), spotted mackerel (*Pneumatophorus tapeinocephalus* Bleeker) (Kawamura, 1976), chum salmon (*Oncorhynchus keta* Walbaum) (Ryer & Olla, 1991; Davis & Olla, 1992), perch (*Perca fluviatilis* L. (Eklöv, 1992) and walleye pollock (Ryer & Olla, 1992). Studies of social facilitation with direct relevance to baits and stock assessment have also been conducted. For example, a clear demonstration of social facilitation in feeding behaviour occurred when Shardlow (1993) used underwater cameras to quantify the number of salmon approaching and attacking lures on trolling gear. As a result of this facilitation, salmon catch rate was not linearly related to density because per capita attacks by salmon (catchability) increased with abundance. A second demonstration of social facilitation directly related to fishing methods is provided by an experimental study conducted in large laboratory pools with sets of one, two and four Pacific halibut (Stoner & Ottmar, 2004). Single fish had lower levels of general activity and responded more slowly to an olfactory cue from squid than

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multiple fish. Also, multiple fish located and consumed baits more quickly than single fish, had a higher per capita attack rate on baits, and a reduced latency period between bait location and strike. Overt aggression was not observed among fish of equal size, and competition for the limited numbers of baits appeared to be exploitative. Therefore, social facilitation in fishes appears to be a function of increased activity, increased collective sensitivity to olfactory and visual cues, and increased propensity to strike baits in the presence of potential competitors. It has also been observed that fish struggling on hooks can stimulate others to attack baits (Fernö *et al.*, 1986; Løkkeborg *et al.*, 1989). Social interactions are most likely to occur in the near-field range around baits or pots, where fish interact visually. High fish density, however, can result in stimulated activity and responsiveness anywhere in the active space. Given all of these observations, fish catchability can be expected to increase with fish density in a non-linear function.

DISCUSSION

VARIATION IN FEEDING AND CATCHABILITY

Catch per unit effort (CPUE) provides a comparable index of population density for a variable of interest only when the probability of catching fish with a unit of effort is constant for that variable (Murphy, 1960; Ricker, 1975; Quinn *et al.*, 1985). In simplest form, the relationship between CPUE (U) and population density or abundance (N) is expressed as $U = qEN$, where E represents fishing effort and q is the catchability coefficient. This review reveals widespread evidence that the assumption of constant catchability is often not valid for baited fishing techniques, which depend upon volitional behaviour by the target species. Direct field observations of responses to baits are relatively rare but it is clear that motivation to attack is variable. For example, in field studies conducted in Norway, only 5% of Atlantic cod observed in the near-field range responded to baits and only 29% of those actually attacked (Løkkeborg *et al.*, 1989). In another investigation, only 1·2% of cod observed on video were captured (He, 1996). Responsiveness was higher in haddock (36% responded and 69% of those responding made attacks) (Løkkeborg *et al.*, 1989) and Pacific halibut (43% of the fish observed made attacks) (Kaimmer, 1999). More importantly, it is evident that responsiveness varies in time, in space, and among individuals depending upon the environment and feeding history of the fish.

The published record shows that activity and feeding by fishes is substantially influenced by a large number of environmental variables. The effects on activity and feeding motivation vary but act through four different basic mechanisms characterized in Fig. 1. Certain variables, most notably water temperature and prey availability, affect metabolic processes which, in turn, regulate both activity and feeding motivation. Other variables impose sensory limitations on fishes. For example, changes in the visual environment created by conditions of light, water turbidity and bottom type can result in changes in general activity, swim speed, feeding capability and feeding motivation. The density and size structure of conspecifics, as well as the presence and abundance of certain other animal

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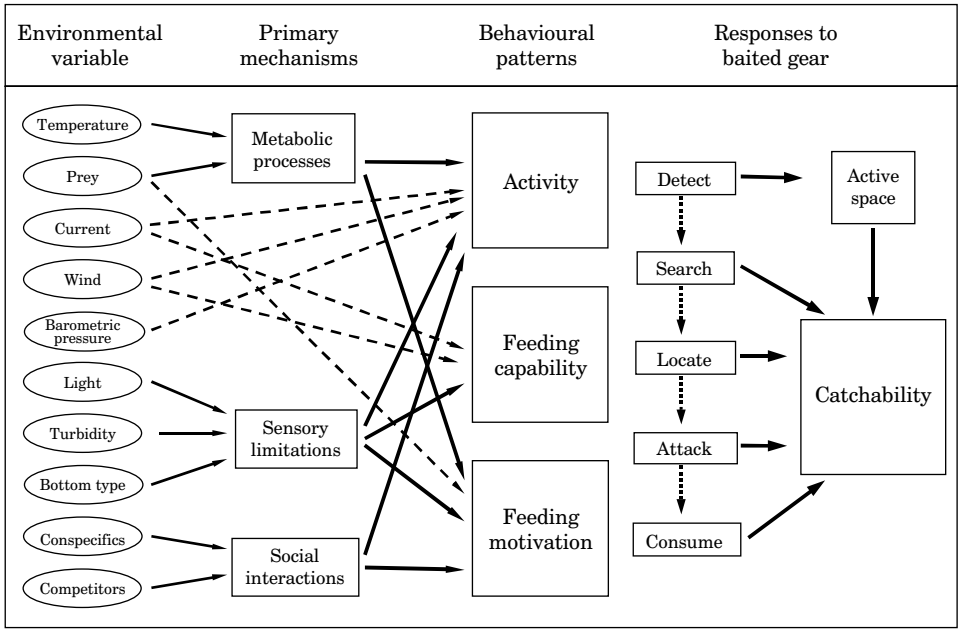

FIG. 1. Environmental variables affect the activity, feeding capabilities and feeding motivation of fishes through several primary mechanisms. In turn, changes in any one of these behavioral patterns can influence the response sequence (*i.e.* bait detection-search-location-attack-consumption) leading to fish capture with baited fishing gear (see text for detailed explanation).

species, result in social interactions including both competition and social facilitation. These interactions affect variation in both activity level and feeding motivation. Finally, certain environmental variables including prey density, current velocity, atmospheric conditions and other lesser understood variables may have direct effects on activity, feeding motivation and feeding capability. These direct effects are shown by the dashed lines in Fig. 1.

Consumption of a bait involves a sequence of behavioural responses normally beginning with the detection of bait scent, followed by systematic searching behaviour, bait location, attack and ingestion (Fig. 1). All of the sequence components are influenced by patterns of fish activity, feeding capability and feeding motivation. For example, changes in activity or swim speed will cause variation in the probability of encountering or detecting a bait chemical plume, the intensity of search and the likelihood that a bait will be located in a given time. Changes in hunger and feeding motivation affect the behavioural threshold for scent detection, and hence the active space created by the bait, as well as search behaviour, bait location rate, and the probability of bait attack and consumption. Changes in visual capability affect search behaviour, bait location and attack. Together these processes regulate both the active space presented by a bait and, ultimately, fish catchability. Because activity, feeding motivation and feeding capability in fishes can all be influenced by a large number of environmental variables, CPUE data from bait-dependent surveys can simply reflect variation in fish catchability rather than the unbiased measures of fish density

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required for estimation of abundance and other population parameters. Therefore, it is critical to understand relationships between environment and feeding biology.

RELATIVE IMPACTS OF ENVIRONMENTAL VARIABLES

With a goal of improving stock assessments, in the face of so many environmental variables that can affect catch rates, it is useful to determine which variables are most likely to have the largest impact on catchability coefficient, and for which variables it will be feasible to adjust stock assessment models. Potential impact of a particular variable depends upon two factors: the natural range of conditions inhabited by the subject species and the strength of its effect on active space and catchability. Feasibility of adjusting a stock assessment model for the impacts of an environmental variable depends upon the ease with which the variable can be measured near the survey gear and knowledge of the relationship to fish catchability. While species-specific complexities necessitate caution in application of corrections to stock assessment models, this review was conducted to identify and rank the most important variables for future consideration (but not to parameterize specific models). Table I was developed using the criteria above to provide general guidance.

Environmental variables evaluated in this study can be placed in the following general categories: 1) variables likely to have strong impacts on overall fish catchability and for which corrections are feasible for fishery stock assessments

TABLE I. Relative characteristics of environmental variables, their potential impacts on fishing with baited gear and stock assessments, and the feasibility of making corrections for environmental variation in stock assessment models. Variation is scored for likelihood that significant spatial or temporal variation in a particular variable will occur within a species' range (see text). Feasibility for corrections is based upon the ease with which the environmental variable can be measured and knowledge of how it effects feeding-related behaviour. All scores range 0 to 3, representing increasing condition (*e.g.* low to high impact)

Environmental variables	Characteristics		Impacts on fishing		Feasibility of correcting models
	Variation	Ease of measurement	Active space	Catchability	
Physical variables					
Temperature	3	3	3	3	3
Light	3	3	1	3	2
Current	3	2	3	2	2
Turbidity	2	1	1	2	1
Bottom type	1	1	1	1	1
Atmospheric conditions	3	–	Unknown	Unknown	0
Biological variables					
Conspecifics	3	1	1	2	2
Prey density	3	1	2	2	1
Other species	3	0	1	2	0

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(temperature is probably the most important variable in this category), 2) variables with high potential for impact but low feasibility for correction and 3) variables likely to have low impact or likelihood for correction.

The potential effects of temperature on fish catchability are profound and adjustments for this variable in stock assessments are particularly urgent. Temperature affects fish activity, swim speed, metabolic processes and feeding motivation. The result is greatly reduced active space and catchability under low temperature conditions. In addition to basic behavioural and physiological observations related to temperature, there is substantial evidence for a significant temperature effect on catchability in both invertebrates and fishes captured with baited fishing gear. For example, catchability coefficients for western rock lobster (*Panulirus cygnus*) in baited pots doubled or tripled with increases in water temperature from 16 to 24° C, and varied widely over several years ($q = 0{\cdot}29$–$1{\cdot}74$) (Chittelborough, 1970; Morgan, 1974). Catchability also varied with lunar cycle. Stoner & Sturm (2004) determined that temperature-related variation in feeding motivation by sablefish would result in catchability on longlines spanning at least one order of magnitude over the observed range of thermal habitat (<2° C to at least 8° C). Fortunately, ectotherm physiology and behaviour follow temperature relationships in relatively predictable patterns (Fry, 1971; Brett, 1979; Angilletta *et al.*, 2002) and temperature is easily measured in the field. Thus, temperature-related variation in q can be relatively easily accommodated into stock abundance models. An example is provided in the section on 'Improving Stock Assessments'.

Light level and current are also likely to have large impacts on fish catchability. Light levels are easily estimated with field recorders. The levels needed for bait location, however, are poorly understood and vary with fish size and age; hence the adjustments required for variable catchability are more difficult to apply than for temperature. Current velocity and direction are also known to have large effects on the active space presented by baits (Løkkeborg *et al.*, 1995). Obtaining measurements is more complicated and expensive than for temperature, but corrections can be made for active space (Stanley *et al.*, 1985; Sainte-Marie & Hargrave, 1987; McQuinn *et al.*, 1988) if care is taken to consider the non-constant capture field (Sigler, 2000). Direct effects of current velocity on fish activity and feeding motivation have not been addressed in stock assessments, but these influences may be less critical than the transport and decay of chemical cues in combination with behavioural response thresholds.

Biological variables, especially the density of conspecifics and prey species, lie primarily within the second general category of variables because they are likely to have large impacts on fish catchability but making accurate corrections to stock assessments is difficult. There are numerous cases where CPUE appears to have a non-linear relationship with stock abundance or stock size (Arreguín-Sánchez, 1996). For example, Peterman & Steer (1981) reported an inverse relationship between the abundance of Chinook salmon (*Oncorhynchus tshawytscha* Walbaum) and catchability on hook and line gear in the sport fishery. The effects of conspecific density, however, are undoubtedly influenced by the scale being considered. For example, while inverse relationships are known to be associated with large-scale distribution of salmon, direct observations on a fine scale (the near-field around baits) reveal that a positive relationship

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can occur because of social facilitation (Shardlow, 1993). Other biological variables such as prey density and the abundance of bait competitors are difficult to assess, and functional relationships to catchability are mostly unknown as they apply to baited gear. Bertrand *et al.* (2002) pointed out the importance of scale in prey distribution, as discussed earlier.

The third category of environmental variables includes turbidity, bottom type and atmospheric conditions. Turbidity is less easy to measure than temperature or light, and relationships with fish feeding biology are poorly understood. Bottom type probably has relatively low impact on catchability and relationships are unknown. Atmospheric conditions, such as barometric pressure and wind, are simple to measure, but impacts and relationships are essentially unknown.

Standardization of fishing methods provides some control over environmental variables affecting catch rates and stock assessments. For example, annual surveys conducted at the same time each year might minimize effects of seasonal variation in temperature and other variables (Engås & Løkkeborg, 1994). Time of day can also be critical. Ward *et al.* (2004) showed recently that catch rates on pelagic longlines are strongly influenced by timing of operations in relation to dawn and dusk (probably an effect of light level) and the effects varied with species. Changes in longline soak times over the course of long-term surveys also created systematic, catchability-related biases in estimates for mortality and abundance, but Ward *et al.* (2004) were able to affect corrections in their models. Internal consistency in catches over time, particularly where individual cohorts can be tracked from year to year, provides some confidence that abundance estimates are robust. Nevertheless, survey standardization alone cannot accommodate significant inter-annual and spatial variation in key environmental factors. The latter will be particularly problematic with fishes such as Atlantic cod, sablefish or Pacific halibut that occupy broad geographic ranges and a wide range of temperature conditions, light levels and prey abundance. For example, fish populations in the coldest or most prey rich sectors of a survey range may be consistently underestimated. A similar spatial bias in stock assessment could occur for demersal fishes that occupy a wide range of depth, with associated variation in temperature, light level, current velocities and prey abundance.

SURVEY COMPARISONS

One way to explore potential biases resulting from environmental variation in catchability is to compare catches on baited gear with results from other types of surveys that depend to a lesser extent on volitional behaviour of the subject species. For example, in one empirical study conducted in British Columbia, abundance estimates based upon hook and line capture of the numerically dominant rockfish (*Sebastes maliger* Jordan & Gilbert) were compared with density estimates made visually from a submersible, where the latter were assumed to represent true density values (Richards & Schnute, 1986). The relationship between CPUE on hooks was directly proportional to fish density when the test was restricted to modal depth for the species and specific weather conditions. The relationship, however, disappeared under more variable

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environmental circumstances, implying variation in fish feeding behaviour. Huse *et al.* (1999) used a combination of trawl, longline and gillnet collections to explore biases in stock assessments for Greenland halibut (*Reinhardtius hippoglossoides* Walbaum). They found that longline gear was selective for large fish because of advantages held by large over small individuals in both exploitative and interference competition. This size-related variation in catchability created biases in life-history parameters derived from the surveys. Longline surveys conducted by the International Pacific Halibut Commission and trawl surveys conducted by the US National Marine Fisheries Service in the Gulf of Alaska show relatively close agreement in long-term, inter-annual trends in CPUE at size for Pacific halibut >80 cm (Clark & Hare, 2003). These data, however, have not been analysed for variation on finer temporal or spatial scales. Biases with depth, temperature and other variables are likely, and future comparisons for important resource species should be evaluated over a range of conditions to test the critical assumption of constant catchability on baited gear. In clear water, visual censuses may be a useful tool for comparison of various sampling methods, but biases are inherent with any survey approach.

IMPROVING STOCK ASSESSMENTS

Ricker (1975) suggested long ago that variability in catchability is likely to be the greatest source of error in stock assessment models based on CPUE where catchability is considered to be constant. There are now a substantial number of investigations that show the sensitivity of stock assessment parameters to variation in active space (Sigler, 2000) or catchability coefficients (Hilborn & Walters, 1992; Arreguín-Sánchez, 1996). Improvements are being made in models to incorporate more sources of variability (Cadrin, 2000; Maunder, 2001; McAllister & Kirchner, 2002; Rahikainen & Kuikka, 2002), particularly those based upon trawl surveys. As yet very few models adjust q to reflect changes in the physical or biological environment other than stock density, and further improvements in modeling are warranted. This is particularly important for stock assessments based upon surveys with baited gear because catch rates depend upon volitional behaviour in the target species and species interactions.

There is at least one good example of how environment-related variation in fish catchability can be incorporated in a stock assessment model. Catchability coefficients employed in stock abundance estimates for yellowfin sole (*Pleuronectes asper* Pallas) in the Bering Sea are now adjusted for temperature because of movements associated with interannual variation in the temperature field and because vulnerability to trawl gear increases with increasing temperature (T. Wilderbuer, pers. comm.). Video recordings made near trawl footropes show that yellowfin sole bury less and herd into trawls more effectively with increasing temperature. Consequently, catchability coefficients for yellowfin sole are adjusted from 0·98 to 1·60 for the observed temperature range of 0·3 to 3·8° C during Bering Sea surveys. Thus, adjustment for temperature in the modeling process can affect a difference of >60% in fish biomass.

Some advances have been made in stock assessment models that use CPUE data from baited fishing gear. Bigelow *et al.* (2002) developed habitat-based models to estimate the relative abundance of Pacific bigeye tuna (*Thunnus*

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obesus Lowe) in terms of effective longline effort instead of simple nominal fishing effort (*e.g.* number of hooks fished). Based upon knowledge of the preferred habitat of bigeye tuna, specifically temperature and dissolved oxygen, their model was combined with oceanographic data to determine what proportion of the fishing gear was within the appropriate depth range and habitat for the target species. This approach accommodates variation in availability but not vulnerability of fish to the gear, and more recent modeling efforts indicate that catchability of pelagic fishes can be variable even within the preferred habitat. When generalized linear mixed models were used to infer depth-dependent catchability of pelagic fishes, catchabilities did not match the distribution of pelagic fish derived from tracking studies (P. Ward, pers. comm.). Additional environmental variables, especially prey density (Bertrand *et al.*, 2002), undoubtedly affect the variation. Others have incorporated model parameters to accommodate variation in catchability as a time function (*e.g.* interannual variation in gear efficiency and diurnal effects) for baited gear (Arreguín-Sánchez & Pitcher, 1999; Ward *et al.*, 2004). Similar modeling techniques will be useful for accommodating environment-related variation in catchability coefficients. More fundamentally, it will be necessary to determine specifically how q is affected by environment.

RECOMMENDATIONS FOR NEW BEHAVIOURAL STUDIES

It is clear from this review that environmental variables can have a significant impact on the feeding behaviour of fishes and produce serious environmental biases in surveys conducted with baited gear. While knowledge of environmental biases relevant to fishery stock assessments is incomplete for most economically important species, new tools are available to make observations on behavioural patterns in fishes that are directly relevant to the science of baited fishing gear. For example, movement patterns and variation in activity can be recorded with acoustic and archival tags, and advances have been made with Atlantic cod (Løkkeborg & Fernö, 1999; Stensholt, 2001; Løkkeborg *et al.*, 2002), certain flatfishes (Metcalfe & Arnold, 1997), salmonids (Walker *et al.*, 2000), tuna (Block *et al.*, 2001) and other species (Nichol & Somerton, 2002; Seitz *et al.*, 2002). Fish tracking will be particularly useful when done in conjunction with continuous recordings of environmental variables or where the tags carry sensors for variables such as temperature, depth and light. Acoustic tags have been coupled with data from current meters to explore the behaviour of Atlantic cod (Arnold *et al.*, 1994; Løkkeborg, 1998). Other researchers have used a combination of ultrasonic tags and acoustic surveys to determine how tuna species move vertically relative to physical conditions, micronektonic prey and pelagic longlines in the tropical Pacific (Josse *et al.*, 1998; Dagorn *et al.*, 2000; Bach *et al.*, 2003).

Direct observations on longline function were made from submersibles more than two decades ago (High, 1980; Grimes *et al.*, 1982). Since then, remotely-operated cameras have been used to record the behaviour of fishes near longlines (Fernö *et al.*, 1986; Kaimmer, 1999; He, 2003) and traps (He, 1993), and video cameras set together with current meters have provided important insights into activity rhythms and responsiveness to bait in Atlantic cod (Løkkeborg

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et al., 1989). Video cameras have limited capabilities in low light conditions and fishes that live in low light or dark conditions often do not behave normally under white light. Illumination with infrared wavelengths, outside the visual capabilities of most fishes, provides for recordings in conditions of darkness (Olla *et al.*, 2000), but usually with a range <2 m.

New high frequency *in situ* imaging sonar systems (acoustic cameras) can provide recordings over larger range than traditional optical cameras, in complete darkness and in highly turbid conditions (Moursund *et al.*, 2003). In some cases the imagery can provide sufficient resolution to distinguish fish species. For example, Dual-Frequency Identification Sonar (DIDSON) has been used to distinguish walleye pollock from salmon entering mid-water trawls (C. Rose, pers. comm.). Also, DIDSON was used in 2004 to observe and record behaviour of Pacific halibut near baited hooks and sablefish in and around fish pots in 200–400 m depth in the northeast Pacific (Rose *et al.*, in prep.). Major fish groups such as sablefish, rockfish, hagfish and different flatfishes were readily distinguished by body form and swimming patterns. Approaches and movement patterns related to current direction and velocity, intraspecific social interactions, exploration of pot tunnels, bait testing and rejection, strikes on hooks, bait competitors and catch losses could all be observed without lights over horizontal distances up to 12 m.

Interactions between stock abundance and potential prey species are more difficult to monitor and evaluate. Acoustic surveys for pelagic fishes and their prey have been conducted in parallel with longline surveys (Bertrand *et al.*, 2002) providing a greater understanding of scale and geographic variation in catch rates, as described earlier. Also, it is now possible to couple catch data with large-scale patterns in water column productivity and surface currents derived from remote-sensing data.

Field studies need to be coupled with experimental analyses directed toward understanding the mechanisms that affect variation in activity patterns, locomotion, sensory capabilities, feeding motivation and search behaviour in economically significant fish species. While laboratory investigations have significant limitations in terms of direct field application (Løkkeborg *et al.*, 1993), this is the only practical way to rigorously control environmental variables and fish feeding histories. Relatively few laboratory experiments have been designed specifically to address issues of bait function (Fernö & Huse, 1983; Johannessen *et al.*, 1993; Løkkeborg *et al.*, 1995; Stoner & Sturm, 2004) and hypothesis-testing experimental work needs to be expanded. Table I helps to identify the environmental variables that need most attention.

CONCLUSIONS

Fish capture rates on baited fishing gear can diverge substantially from direct, linear relationships with fish abundance because the gear depends upon volitional behaviour of the target species and because of the profound effects of certain environmental variables on fish activity, thresholds for response to bait cues, feeding motivation and sensory capabilities. At least three forms of information are needed to improve the accuracy of stock assessments based upon baited gear: 1) It will be critical to quantify relationships between the most

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influential environmental variables and the activity and feeding behaviour of fishes being surveyed, 2) key environmental variables will need to be measured in conjunction with stock assessment surveys at spatial and temporal scales relevant to gear function. Temperature, light and current appear to be among the most critical variables for temperate and sub-arctic surveys and 3) for highest value in fisheries management, stock assessment models will need to accommodate variation in catchability coefficients caused by spatial and temporal variation in environmental variables. Otherwise, catch per unit effort will only serve to estimate the abundance of the target species seriously confounded by spatial and temporal variation in the feeding ecology of that species and its competitors.

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